

Number of Parameter Calculation

$$\left(\begin{array}{c} \text{Shape of width} \\ \text{of filter} \end{array} \times \begin{array}{c} \text{Shape of height} \\ \text{of filter} \end{array} \times \begin{array}{c} \text{number of filter} \\ \text{in previous} \\ \text{layer} \end{array} + 1 \right) =$$

$\times$ number of filter
in this layer.

∴ if there is no previous layer filter used
then
instead of used $\left[\begin{array}{c} \text{number of filter} \\ \text{in previous layer} \end{array} \right]$ $\rightarrow$ $\left[\begin{array}{c} \text{use number of} \\ \text{channel} \end{array} \right]$

∴ input-layer "0" parameter update

∴ activation "0" parameter update

∴ pooling layer "0" " " "

∴ Fully connected layer: number of parameter
= $\left(\begin{array}{c} \text{current layer} \\ \text{neuron} \end{array} \times \begin{array}{c} \text{previous layer} \\ \text{neuron} \end{array} \right) + \begin{array}{c} \text{current layer} \\ \text{neuron} \end{array}$

Parameters in CNN

e.g 1

Parameters

• myinput = Input + shape = (10, 10, 1)

0

• myoutput = Convolution 2D (1, 3, 3) (my-input)

10

$$\begin{aligned} & ((10-1)+1) \\ & ((10-3)+1) = 8 \end{aligned}$$

↓

(1, 8x8)



∴ (shape of width of filter × shape of height of filter × number of filter in previous layer + 1) × number of filter in this layer

no of channel

$$(3 \times 3 \times 1 + 1) \times 1 = 10$$

• my-output = Activation('sigmoid') (my-output)

0

• model = Model(myinput, my-output)

total

10

ex 2

- my-input = Input (shape = (10, 10, 1))
- my-output = Convolution2D(5, 3, 3) my-input

channel

parameter

0

50

$$\begin{aligned} &\Downarrow \\ &[(i-k)+1] \\ &[(10-3)+1] = 8 \end{aligned}$$

$$\text{output shape} \# (5, 8 \times 8)$$

Shape of width of filter $\times$ shape of height of filter $\times$ number of filter in previous layer $+ 1$ $\times$ number of filter in this layer.

number of channel

$$\begin{aligned} &= (3 \times 3 \times 1 + 1) \times 5 \\ &= 10 \times 5 = 50 \end{aligned}$$

- my-output = Activation('sigmoid')(my-output)
- my-output = Convolution2D(2, 3, 3)(my-output)

0

92

$$[(i-k)+1] = 6$$

$$\text{output shape} \# (2, 6 \times 6)$$

Shape of width of filter $\times$ shape of height of filter $\times$ number of filter in previous layer $+ 1$ $\times$ number of filter in this layer.

$$\begin{aligned} &(3 \times 3 \times 5 + 1) \times 2 \\ &46 \times 2 = 92 \end{aligned}$$

- my-output = Activation('sigmoid')(my-output)
- model = Model(my-input, my-output)

0

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e.g 3Parameter

- my_input = Input (shape = (100, 100, 3))
- my_output = Convolution2D(8, 3, 3)(my_input)

$$(i - k) + 1$$

$$(100 - 3) + 1 = 98$$

output shape # (8, 98x98)

$$\left(\begin{array}{c} \text{shape of} \\ \text{width of} \\ \text{filter} \end{array} \times \begin{array}{c} \text{shape of} \\ \text{height of} \\ \text{filter} \end{array} \times \begin{array}{c} \text{number of} \\ \text{filter in} \\ \text{previous} \\ \text{layer} \end{array} + 1 \right) \times \begin{array}{c} \text{number of} \\ \text{filter in} \\ \text{this layer} \end{array}$$

↓
of used channel.

$$(3 \times 3 \times 3 + 1) \times 8$$

$$28 \times 8 = 224$$

- my_output = Activation('sigmoid')(my_output) 0
- my_output = Convolution(1, 3, 3)(my_output) 73

$$= [98 - 3] + 1$$

$$= 96$$

output shape # (1, 96x96)

$$\left(\begin{array}{c} \text{filter} \\ \text{width} \end{array} \times \begin{array}{c} \text{filter} \\ \text{height} \end{array} \times \begin{array}{c} \text{number of} \\ \text{filter in} \\ \text{previous} \\ \text{layer} \end{array} + 1 \right) \times \begin{array}{c} \text{number of filter} \\ \text{in this layer} \end{array}$$

$$(3 \times 3 \times 8 + 1) \times 1 = 73$$

- my_output = Activation('sigmoid')(my_output) 0
- model = Model(my_input, my_output)

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e.g 4

→ my_input = Input (shape = (100, 5)) 0
width, no height
channel

→ my_output = Convolution1D (8, 3) 128
filter

$$(100 - 3) + 1 = 98$$

$$\Downarrow$$
$$(8, 98)$$

filter

$$\left(\begin{array}{c} \text{filter} \\ \text{width} \end{array} \times \begin{array}{c} \text{number of} \\ \text{channel} \end{array} + 1 \right) \times \begin{array}{c} \text{number of} \\ \text{filter widths} \\ \text{layer} \end{array}$$

$$(3 \times 5 + 1) \times 8 = 128$$

→ my_output = Activation('sigmoid')(my_output) 0

→ my_output = Convolution1D (1, 3)(my_output) 25

$$(98 - 3) + 1 = 96$$

$$\text{output } (1, 96)$$

$$\left(\begin{array}{c} \text{filter} \\ \text{width} \end{array} \times \begin{array}{c} \text{number of filter} \\ \text{in previous} \\ \text{layer} \end{array} + 1 \right) \times \begin{array}{c} \text{number of} \\ \text{layer in} \\ \text{this layer} \end{array}$$

$$(3 \times 8 + 1) \times 1 = 25$$

→ my_output = Activation('sigmoid')(my_output) 0

→ model = Model (my_input, my_output) 0
153

e.g.

Output - slope

paramita paramita

→ Conv2d (32, 3, 3) 26, 26, 32 Bia. 320

$$(3 \times 3 \times 1 + 1) \times 32 = 320$$

→ mapping to	13, 13, 32	16	0
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→ $\text{Conv } 2d-1(32 \times 3 \times 3) \quad (11, 11, 32)$
 $[13-3] + 1 =$

$$\left(\begin{array}{c} \text{filler} \\ \text{written} \end{array} \times \begin{array}{c} \text{filler} \\ \text{height} \end{array} \times \begin{array}{c} \text{\# of filler} \\ \text{in previous} \\ \text{layer} \end{array} + 1 \right) \times \begin{array}{c} \text{number of} \\ \text{filler} \\ \text{in this} \\ \text{layer} \end{array}$$

$$(3 \times 3 \times 32 + 1) \times 32$$
$$289 \times 32 = 9248$$

→ max pollup2d-1 (5, 5, 32)

→	flatter the	(Nov, 800)	0
		5x5A32 = 800	

→ `dense(128)` (None, 128)

$$\left(\begin{array}{c} \text{curr layer} \\ \text{neuron} \end{array} \times \begin{array}{c} \text{previous layer} \\ \text{neuron} \end{array} \right) + \begin{array}{c} \text{curr layer} \\ \text{neuron} \end{array}$$

$$(128 \times 800) + 228 = 103528$$

→ dense_1(64) (None, 64)

$$(64 \times 128) + 64 = 8256$$

↓ ↓ ↓
 current layer previous current
 layer layer layer
 neuron neuron neuron

→ dense-1(10)	(None, 10)	650
	$(10 \times 64) + 10 = 650$	

121002

dense layer is also known as
fully connected layer.

parameter & hyperparameter
in CNN

hyperparameter

kernel-size

stride

padding

pooling

number of filters

parameter

weight

bias

other parameter

→ Activation function parameter

→ Normalization parameter

↳ batch normalization

↳ layer normalization

→ Drop-out parameter

→ fully connected layer parameter

→ loss function parameter

Fully Connected layer / Dense layer

layers = Dense (units = 200 , activation = 'sigmoid',

↓
number of
neuron

input_shape = (748, 748)

Dropout (rate = 0.5)

parameters :-

(# of neurons in curr layer × # of neurons in previous layer) + # of neurons in curr layer.

↳ Pooling layer

Hyperparameters :-

↳ pooling type → max
→ average

↳ pool size (kernel-size)

↳ stride (by default same as pool size)

↳ padding

parameters
NO

↳ Convolution layer :-

Hyperparameters

→ kernel-size
→ stride (2)
→ padding
→ Number of filter
→ Activation function

parameters

= (filter width × filter height × # of filter in previous layer / channel + 1) × no of filter in curr layer.